Alireza Nasirianfar

(+1) (438) 509-6390 • alireza.nasirianfar@umontreal.ca • alireza-nasirian
alireza-nasirian

Education

Université de Montréal **Montreal, Canada**
M.Sc. Computer Science *2026 – Present*
Specialization: Software Engineering
GPA: 4.3/4.3. Selected coursework: Art algorithmique et logiciel libre, Deep Learning for NLP. (Transcript)

Amirkabir University of Technology (Tehran Polytechnic) **Tehran, Iran**
B.Sc. Computer Engineering *2020 – 2025*
GPA: 17.72/20. Selected coursework: Artificial Intelligence, Information Retrieval, Algorithms, Data Structures, Linear Algebra. (Transcript)

Research Interests

- Software Supply Chain Security • Empirical Software Engineering
- CI/CD Security • Static Analysis • Software Engineering Automation

Language Skills

- Persian: Native
- English: Academic IELTS Test 7/9 (Listening:7.5, Reading:7.5, Speaking:7, Writing:6)

Teaching Experience

- Winter 2026 **Algorithmic Art and Free Software**, TA (Supervisor: Prof. Benoit Baudry) UdeM
- Spring 2024 **Information Retrieval**, TA (Supervisor: Dr. Nickabadi) AUT
- Spring 2024 **Computer Networks**, TA (Supervisor: Dr. Sabaei) AUT
- Spring 2024 **Software Engineering II**, TA (Supervisor: Dr. Gohari) AUT
- Spring 2024 **Assembly Language**, TA (Supervisor: Dr. Farbeh) AUT
- Fall 2023 **Advanced Programming**, TA (Supervisor: Dr. Ziaeetabar) AUT

Research Experience

Université de Montréal (UdeM) **Montreal, Canada**
Research Assistant *Jan. 2026 – Present*
Supervisor: Prof. Benoit Baudry
- Conduct empirical research on software supply chain security, with an emphasis on vulnerabilities in GitHub Actions and CI/CD workflows.
- Developed WAVES, a manually verified benchmark of 475 workflows containing 2,424 evidence-linked security weakness instances across 10 categories.
- Designed a diversity-aware sampling approach over a corpus of approximately 1.05 million workflow records, considering structural complexity, trigger risk, third-party action dependencies, and security exposure.
- Evaluated GitHub Actions security scanners, including Poutine and Zizmor, to characterize detection coverage and identify gaps between tool findings and manually validated weaknesses.

Amirkabir University of Technology **Tehran, Iran**
Research Assistant *Dec. 2023 – Apr. 2025*
Developed a Persian social-media stance detection system under the supervision of Dr. Nickabadi, using semi-supervised learning and transformer-based language models.

Work Experience

BoostSecurity (Website) **Montreal, Canada**
Security Research Intern *May 2026 – Sep. 2026*

- Investigated security weaknesses in real-world GitHub Actions workflows and CI/CD pipelines using static-analysis techniques.
- Contributed to the construction and manual validation of the WAVES benchmark for evaluating workflow security scanners.
- Refined vulnerability-labeling procedures and analyzed scanner findings to support reproducible security evaluations.
- Collaborated with academic and industry researchers on benchmark methodology and workflow security assessment.

Tosan (Website) **Tehran, Iran**
Software Engineer – Payment Team *Feb. 2022 – Dec. 2025*

- Developed microservice-based backend systems integrating REST and SOAP services.
- Built reactive APIs using Spring Reactor and Spring WebClient.
- Implemented authentication and authorization using OAuth 2.0, Spring Security, and Keycloak.
- Worked on high-volume banking and ISO 8583 transaction systems using Oracle Database and jPOS.

Selected Projects

- **WAVES – GitHub Actions Security Benchmark:** Developed a benchmark of 475 real-world workflows with 2,424 manually verified security weakness instances across 10 categories. Used it to evaluate Poutine and Zizmor, analyze rule coverage and missed detections, and measure scanner performance (Zizmor: 0.70 precision, 0.42 recall).
- **Persian Social-Media Stance Detection:** Developed a stance detection pipeline for Persian tweets using text preprocessing, semi-supervised label propagation (KNN and RBF), and transformer-based models including ParsBERT and SentenceBERT. *B.Sc. Capstone*
- **Rating App:** Built a customer-feedback REST API with Spring Boot, Hibernate, Spring Security, JUnit, and Mockito. (GitHub)
- **Image Classification:** Implemented MLP and CNN models in PyTorch for CIFAR-10 image classification, achieving 50% and 70% test accuracy, respectively. (GitHub)
- **Search Engine:** Built an information retrieval engine with normalization, stemming, and a positional inverted index. (GitHub)

Technical Skills

- **Programming:** Python, Java, Kotlin, C, SQL
- **Security and CI/CD:** GitHub Actions, Poutine, Zizmor, CI/CD security, static analysis, Git, Docker
- **Backend Engineering:** Spring Boot, Spring WebFlux/Reactor, Spring Security, REST, SOAP, microservices, OAuth 2.0, Keycloak
- **Data and Research:** BigQuery, Pandas, Scikit-learn, benchmark design, empirical evaluation
- **Databases and ML:** Oracle Database, Redis, Elasticsearch, PyTorch, TensorFlow, Hugging Face